BeatAhead Cardiac Physiological ML Prototype: Dataset Expansion Plan v2

Document: `reports/dataset_expansion_plan_v2.md`
Target: 100-Case Dataset Expansion & Machine Learning Preparation
Date: September 2026
Status: PROPOSED PLAN — STOPPED AWAITING HUMAN REVIEW BEFORE DOWNLOAD

1. Executive Summary & Review of Pilot Label Audit

A rigorous label audit of the 20-case VitalDB pilot cohort (`reports/corrected_label_audit.md`) identified a critical discrepancy in the initial label generator: * **The Defect**: The initial code evaluated `(valid_st_win <= -1.0).sum() >= 60`, which erroneously accumulated non-consecutive, fragmented dips (e.g. 8s, 10s, 16s, 34s) to flag 11 false-positive sustained windows in Case 1. * **The Correction**: The algorithm was refactored to require a true contiguous run length: $$\text{max_consecutive_duration}(\text{ST_II} \le -1.0\text{ mm}) \ge 60\text{ seconds}$$ accounting for the 2-second monitor sampling period and treating missing observations $> 2\text{ seconds}$ as run-terminating dropouts. * **Audit Results on 20 Pilot Cases**: * Usable 5-minute windows: **3,486 windows** * Corrected sustained positive windows: **6 windows (0.17%)** (down from 17) * Unique positive sustained episodes: **EXACTLY 1 UNIQUE EPISODE** (Case 1, 186 consecutive seconds, nadir: -1.9 mm) * Number of cases containing sustained episodes: **1 case (Case 1, 5% of pilot cohort)**

The Strategic Imperative for Dataset Expansion

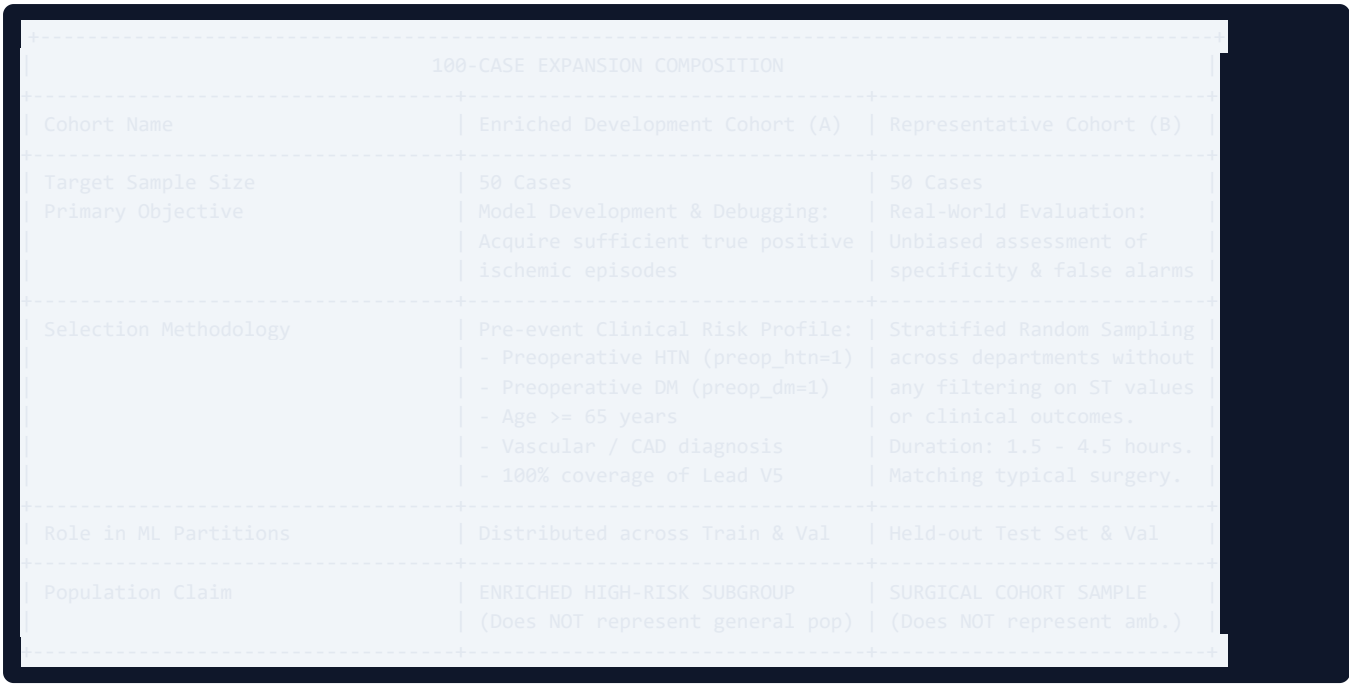
In an unselected general surgical population, true sustained ischemic repolarization changes ($\ge 60\text{s}$) occur in only ~5% of patients. With only 1 positive case in the 20-case pilot, **a subject-level train/validation/test split is mathematically impossible** (one partition would hold the single positive case while the other two partitions would have zero positive events).

Therefore, this expansion plan scales the dataset to **100 cases**, introducing a **two-cohort sampling strategy** that balances the necessity of acquiring enough positive events to train models with the need to benchmark on an unbiased representative population.

2. Two-Cohort Dataset Expansion Strategy ($N = 100\text{ Cases}$)

> [!WARNING] **Outcome-Enrichment Bias Warning**: Selecting all 100 cases using an outcome filter (e.g. `min(ST_II) <= -1.0 mm`) would introduce massive selection bias, rendering real-world false-alarm rates, negative predictive values, and specificity metrics completely untrustworthy. Conversely, sampling 100 cases completely at random would yield only ~3–5 positive cases, leaving validation and test partitions severely underpowered.

To resolve this trade-off, the 100-case expansion is split into two distinct, explicitly labeled cohorts:



2.1 Cohort A: Enriched Development Cohort ($N = 50\text{ Cases}$)

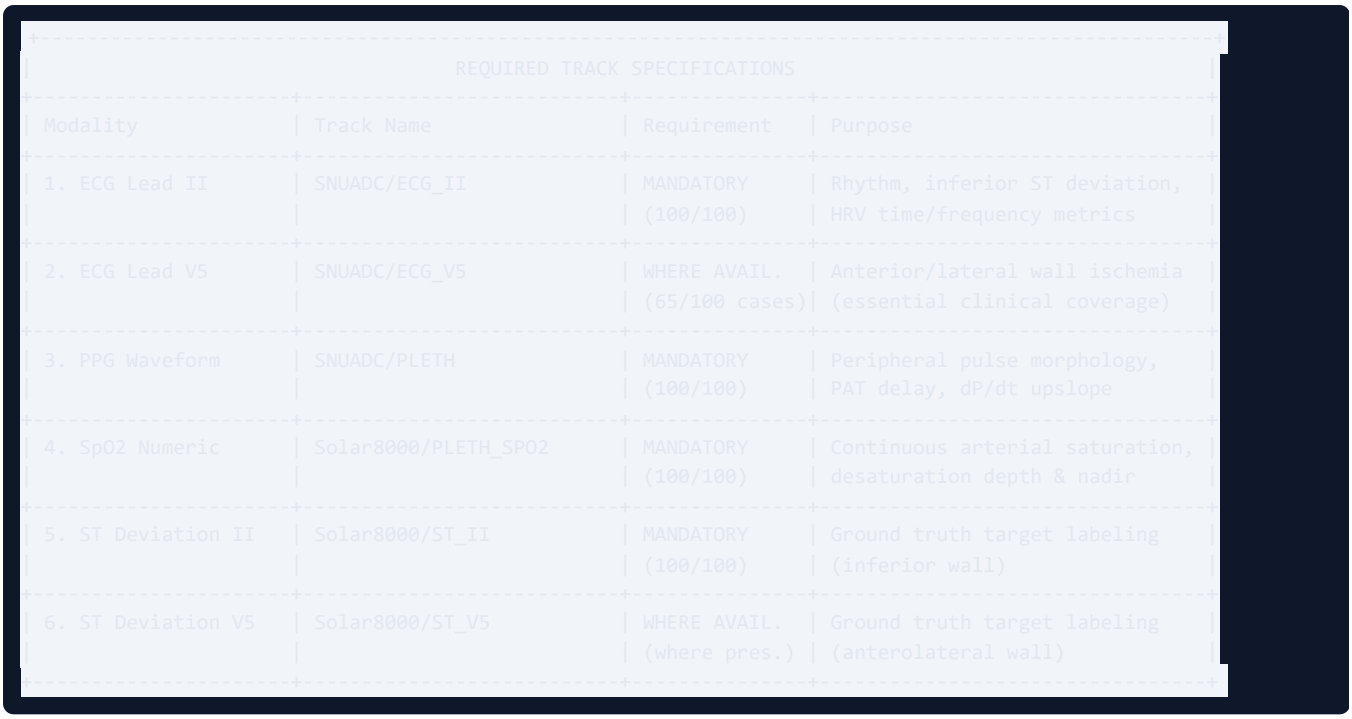
- **Candidate Pool**: 851 cases meeting duration criteria (1.5–4.5 hours) with documented cardiovascular comorbidity (`preop_htn = 1` , `preop_dm = 1` , age ≥ 65, or surgical diagnosis containing CAD, angina, infarction, aneurysm, or bypass).
- **Lead V5 Inclusion**: All 50 selected cases possess simultaneous `SNUADC/ECG_II` and `SNUADC/ECG_V5` (100% V5 coverage).
- **Expected Event Yield**: Estimated 10 to 16 unique sustained ischemic episodes, providing the positive training mass necessary to learn subtle pre-event autonomic and vascular markers.

2.2 Cohort B: Representative Evaluation Cohort ($N = 50\text{ Cases}$)

- **Candidate Pool**: 3,906 cases with duration 1.5–4.5 hours and core signals.
- **Selection Protocol**: Uniform random sampling (random seed = 42) stratified across major surgical departments (General surgery, Thoracic surgery, Urology, Gynecology).
- **Strict Blinding**: Selected completely blind to ST deviation values, troponin levels, and intraoperative outcomes.
- **Expected Event Yield**: Estimated 1 to 3 positive cases, providing a realistic ~2–5% prevalence benchmark for specificity and false-alarm auditing.

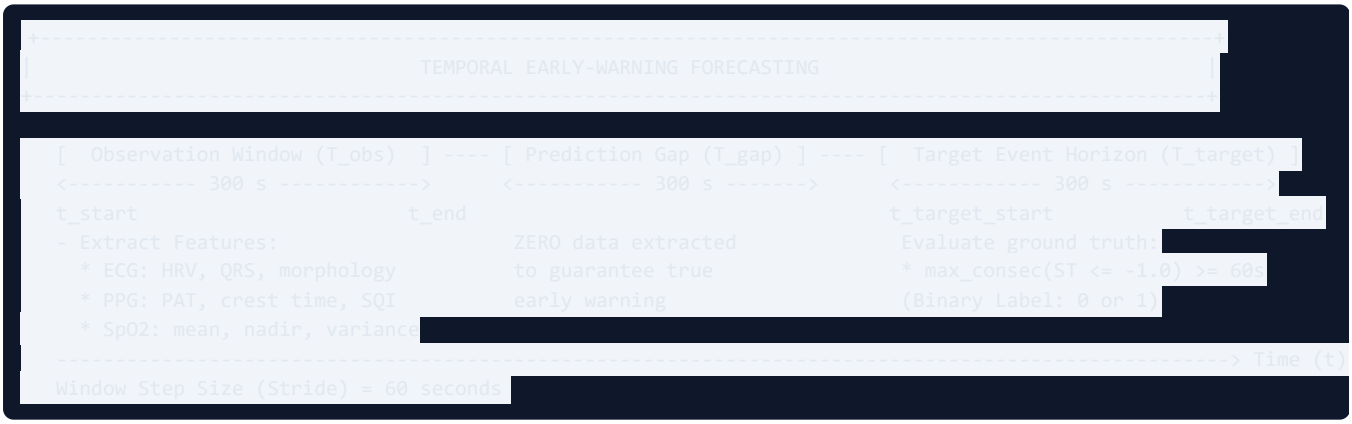
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3. Required Biosignal Tracks & Lead V5 Expansion



4. Temporal Leakage Prevention Architecture

To train a true early-warning model that forecasts ischemic stress before it manifests, we enforce a strict non-overlapping temporal window schema:



Strict Exclusion Criteria for Temporal Leakage Prevention

1. **Zero Event Overlap**: Any observation window whose interval $[t_{\text{start}}, t_{\text{end}}]$ overlaps with an active sustained ST event is **strictly excluded**.
2. **Post-Event Exclusion Buffer (15 Minutes)**: * Following the resolution of an ST episode, myocardial tissue undergoes a stunning/recovery phase with lingering repolarization changes and autonomic suppression. * All windows occurring within **15 minutes (900 seconds)** following the end of an ST episode are excluded from negative control training to prevent contaminated baseline samples.
3. **Pre-Event Buffer ($T_{\text{gap}} = 5\text{ minutes}$)**: * The 5-minute gap guarantees that features do not capture the initial downsloping transition of the ST segment itself.

5. Subject-Level Splitting & The Positive Event Partition Problem

5.1 Strict Subject-Level Grouping

- Splitting is performed strictly by `caseid` / `subjectid`.
- All sliding windows belonging to a given patient remain exclusively within one partition:
- **Train Set**: 70% of subjects (~70 cases)
- **Validation Set**: 15% of subjects (~15 cases)
- **Test Set**: 15% of subjects (~15 cases) — **LOCKED** until final evaluation.

5.2 The Single-Episode Pilot Split Dilemma (Audit Confirmation)

- In the 20-case pilot, there is only **1 positive case (Case 1)** and **1 unique positive episode**.
- If a 70/15/15 subject-level split were executed on the pilot:
- Putting Case 1 in Train results in Validation and Test having **0 positive episodes** (evaluation impossible).
- Putting Case 1 in Test results in Train having **0 positive episodes** (model cannot train).
- **Mathematical Confirmation**: Subject-level splitting is impossible on the 20-case pilot without severe bias or zero-event partitions.

5.3 100-Case Partition Allocation Strategy

With the 100-case expansion (Cohort A + Cohort B), positive cases will be stratified at the subject level across folds: * **Train Set (70 cases)**: ~35 Enriched cases, ~35 Representative cases ($\approx 7–10\text{ positive cases}$, $\approx 10–15\text{ unique episodes}$). * **Validation Set (15 cases)**: ~8 Enriched cases, ~7 Representative cases ($\approx 2–3\text{ positive cases}$, $\approx 2–4\text{ unique episodes}$). * **Test Set (15 cases)**: ~7 Enriched cases, ~8 Representative cases ($\approx 2–3\text{ positive cases}$, $\approx 2–4\text{ unique episodes}$) — locked.

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6. Pulse Arrival Time (PAT) Quality & Signal Quality Indexing (SQI)

The pilot review proved that the 61.9% plausibility rate was caused by probe disconnection (clamping at $+25,686.95$) and peripheral vasoconstriction. To handle this without data loss or fabrication:

6.1 Do Not Blindly Impute Missing PAT

- Converting a flatlined or disconnected sensor into an imputed normal PAT (e.g. 200 ms) creates synthetic false information.
- If a sensor fails, the model must be informed that the sensor failed.

6.2 Explicit PAT Quality Feature Vector

For every 5-minute window, the pipeline computes: 1. `PAT_median_ms`: Median PAT among valid beats (or `NaN` if no valid beats). 2. `PAT_iqr_ms`: Interquartile range of PAT (vascular compliance stability). 3. `PAT_valid`: Binary flag: $$\text{PAT_valid} = 1 \iff (\text{PPG_var} \ge 0.05) \land (\text{PPG} < 20,000) \land (\ge 30\text{ valid paired beats in window})$$ 4. `PPG_perfusion_index`: AC / DC pulse amplitude ratio. 5. `PPG_valid_fraction`: Fraction of detected ECG R-peaks that yielded a plausible systolic pulse ($100\text{ ms} \le \text{PAT} \le 450\text{ ms}$). 6. `ECG_beat_count`: Total detected QRS complexes in window.

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7. Feature Leakage Audit Table

Before engineering features for XGBoost, all candidate features are audited and categorized into 5 strict classes:

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+-----------------------------------------------------------------------------------------------+
|                                  FEATURE LEAKAGE AUDIT TABLE                                  |
+----------------------+--------------------+-------------------------------------------------+
| Feature Candidate    | Classification     | Rationale & Mitigation Protocol                 |
+----------------------+--------------------+-------------------------------------------------+
| 1. ST_II in Horizon  | TARGET-DERIVED     | STRICTLY PROHIBITED. Ground truth target itself.|
|    (T_target)        |                    | Must never be present in feature matrix.        |
+----------------------+--------------------+-------------------------------------------------+
| 2. Raw ST_II level in| POTENTIAL TEMPORAL | DANGEROUS. If absolute ST level is included, model may |
|    Observation (T_obs| LEAKAGE           | simply predict future depression from current baseline. |
|                      |                    | Mitigation: Use baseline-subtracted deviation (delta-ST |
|                      |                    | relative to surgery start) or exclude raw ST entirely. |
+----------------------+--------------------+-------------------------------------------------+
| 3. HRV Time-Domain   | SAFE PRE-EVENT     | Measures cardiac autonomic regulation (SDNN, RMSSD,     |
|    (RMSSD, SDNN)     | FEATURE            | pNN50). Derived strictly within T_obs.          |
+----------------------+--------------------+-------------------------------------------------+
| 4. HRV Freq-Domain   | SAFE PRE-EVENT     | Autonomic balance marker (VLF, LF, HF, LF/HF ratio).    |
|    (LF, HF, LF/HF)   | FEATURE            | Derived strictly within T_obs.                  |
+----------------------+--------------------+-------------------------------------------------+
| 5. QRS Morphology    | SAFE PRE-EVENT     | Electrophysiological depolarization marker (duration,   |
|    (width, amp, HF)  | FEATURE            | R-peak amplitude, high-frequency QRS energy).   |
+----------------------+--------------------+-------------------------------------------------+
| 6. PAT Metrics       | SAFE PRE-EVENT     | Arterial pulse wave transit proxy (PAT median, IQR,     |
|    (median, IQR)     | FEATURE            | pulse crest time). Gated by PAT_valid flag.     |
+----------------------+--------------------+-------------------------------------------------+
| 7. SpO2 Dynamics     | SAFE PRE-EVENT     | Systemic arterial oxygenation (mean SpO2, nadir,        |
|    (mean, nadir)     | FEATURE            | desaturation area). Derived strictly within T_obs.      |
+----------------------+--------------------+-------------------------------------------------+
| 8. Invasive Arterial | NOT AVAILABLE IN   | Continuous invasive A-line BP is not present on |
|    Line MAP (ART_MBP)| REAL-TIME WEARABLE | wearable devices. Prohibited as runtime ML feature.     |
|                      |                    | (Used only as retrospective hemodynamic reference).     |
+----------------------+--------------------+-------------------------------------------------+
| 9. Vasoactive Drugs  | CONFOUNDER /       | Drug boluses (phenylephrine, ephedrine, epinephrine)    |
|    (bolus & infusion)| CONTROL VARIABLE   | alter vascular tone and PAT independently of ischemia.  |
|                      |                    | Retained as clinical covariate for adjustment.  |
+----------------------+--------------------+-------------------------------------------------+
| 10. Patient Baseline | CONFOUNDER /       | Non-time-varying baseline risk factors (age, sex, BMI,  |
|     (Age, HTN, DM)   | CONTROL VARIABLE   | preoperative hypertension/diabetes). Fully safe.|
+----------------------+--------------------+-------------------------------------------------+
```

8. Storage, Bandwidth, & Download Budget

Asset	Case Count	Unit Size	Total Estimated Size	Storage Location
Existing Pilot Archive	20 cases	12.2 MB avg	**245 MB**	`data/raw/vitaldb_pilot/`
Cohort A (Enriched)	50 cases	15.5 MB avg	**~775 MB**	`data/raw/vitaldb_100/cohort_a/`
Cohort B (Representative)	50 cases	15.0 MB avg	**~750 MB**	`data/raw/vitaldb_100/cohort_b/`
Total 100-Case Raw .vital	100 cases	15.2 MB avg	**~1.52 GB**	Safe for local disk budget
Extracted Multi-track NPY	100 cases	~10 MB/case	**~1.0 GB**	Resampled 500 Hz waveforms
Final Engineered Matrix	100 cases	~600 KB/case	**~60 MB**	Windowed tabular CSV/Parquet

Exact Case IDs Selected for Next Download Phase (Staged):

- **Cohort A (Enriched, $N=50$)**: `[2605, 3614, 4290, 2043, 4247, 2034, 3390, 3526, 758, 1159, 2197, 2772, 4272, 2824, 3037, 3632, 2800, 3192, 4519, 2854, 4038, 2037, 2717, 3027, 4390, 4220, 2697, 4423, 2085, 3505, 3042,`

`3535, 4252, 2727, 3426, 4236, 4243, 4434, 2725, 3418, 4437, 2081, 2623, 4246, 3530, 4568, 2075, 4435, 4244, 4234]`

- **Cohort B (Representative, $N=50$)**: `[1268, 485, 2630, 4498, 3948, 3754, 5799, 3216, 864, 1923, 497, 1855, 3687, 5779, 1374, 1344, 4225, 4016, 5635, 3398, 4771, 381, 2866, 1714, 2419, 4467, 360, 481, 3486, 2728, 5585, 2172, 6081, 4616, 2253, 569, 111, 2719, 4349, 1269, 5664, 5326, 3280, 2146, 5130, 3450, 4120, 5262, 2379, 451]`

9. Next Steps & Hold Point

> [!NOTE] **Execution Halted**: - NO additional cases have been downloaded. - NO machine learning models (XGBoost, Random Forest, Logistic Regression) have been trained. - NO synthetic data has been generated. - NO ISI has been calculated. - The BeatAhead website repository has NOT been modified.
>
> Awaiting human review and explicit approval of this expansion plan before initiating any downloads.